ECOM90009 Quantitative Methods for Business

Workshop 5 (Week 6)

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
- Ph.D student in Economics at Unimelb

If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Question 1

A random variable X has a t distribution with 6 degrees of freedom. The random variable Z has a standard normal distribution.

Indicate whether each statement below is true or false. Justify your answer.

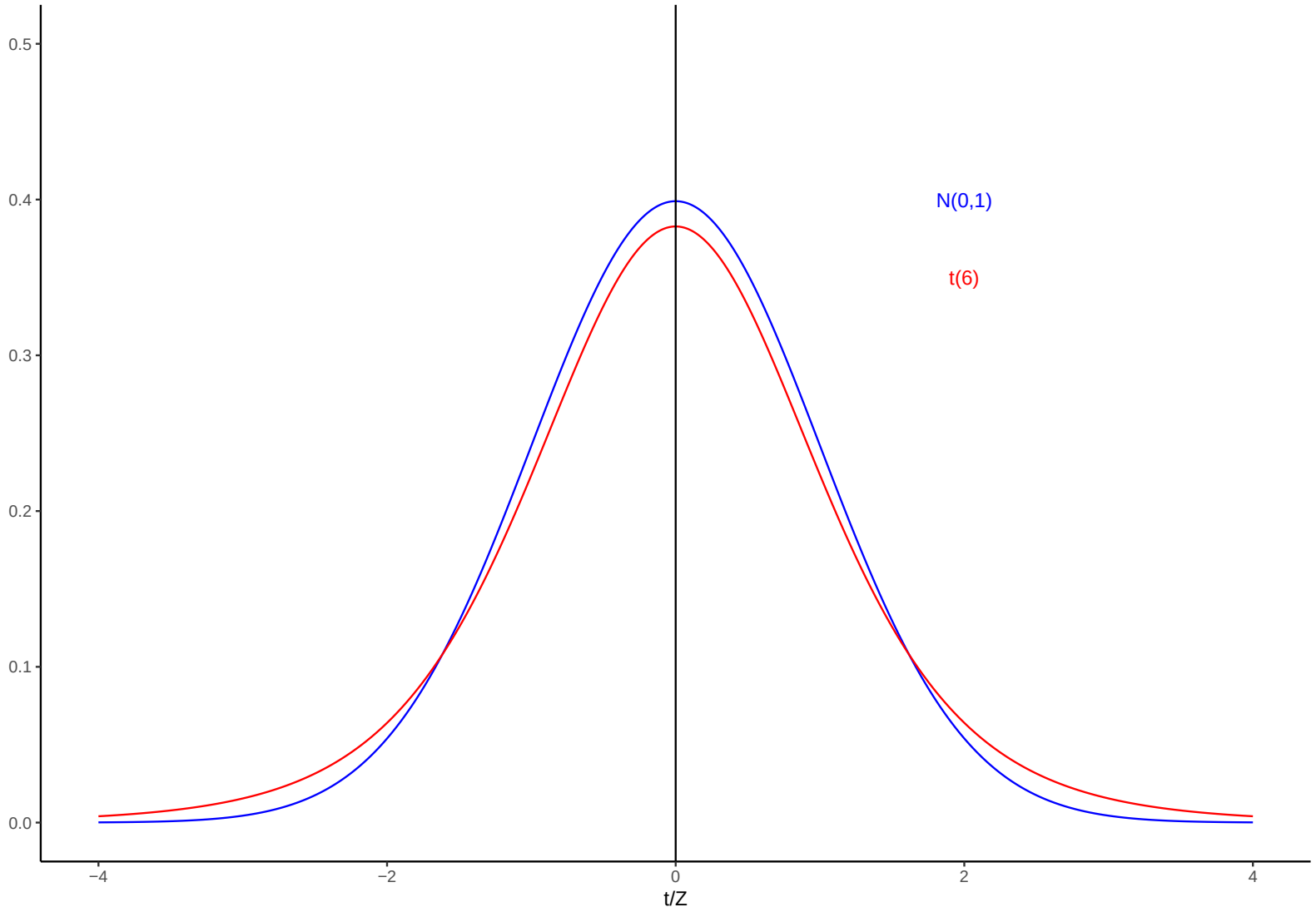
a. $E[X] = E[Z]$
b. $V[X] < V[Z]$
c. $P(X > 0) = 0.5$
d. $P(X > 2) > P(Z > 2)$

Question 1(a)

A random variable X has a t distribution with 6 degrees of freedom. The random variable Z has a standard normal distribution.

Statement: $E[X] = E[Z]$

Comparison of $t(6)$ and $N(0,1)$



Question 1(b)

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Statement: $V[X] < V[Z]$

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- From p.392 of the textbook, the mean and variance of a t random variable are:

$$E(t) = 0, \qquad V(t) = \frac{n-1}{n-3} \text{ for } n > 3$$

- As $(n-1) > (n-3)$, $V(t)$ is always greater than 1.

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- As $(n-1) > (n-3)$, $V(t)$ is always greater than 1.
- The t distribution is more widely dispersed than the standard normal distribution.
  - Variance of a standard normal distribution is 1.
- In the application of estimating the population mean μ when population variance σ^2 is unknown, we use the t distribution with degrees of freedom $\nu = n - 1$.

Question 1(c)

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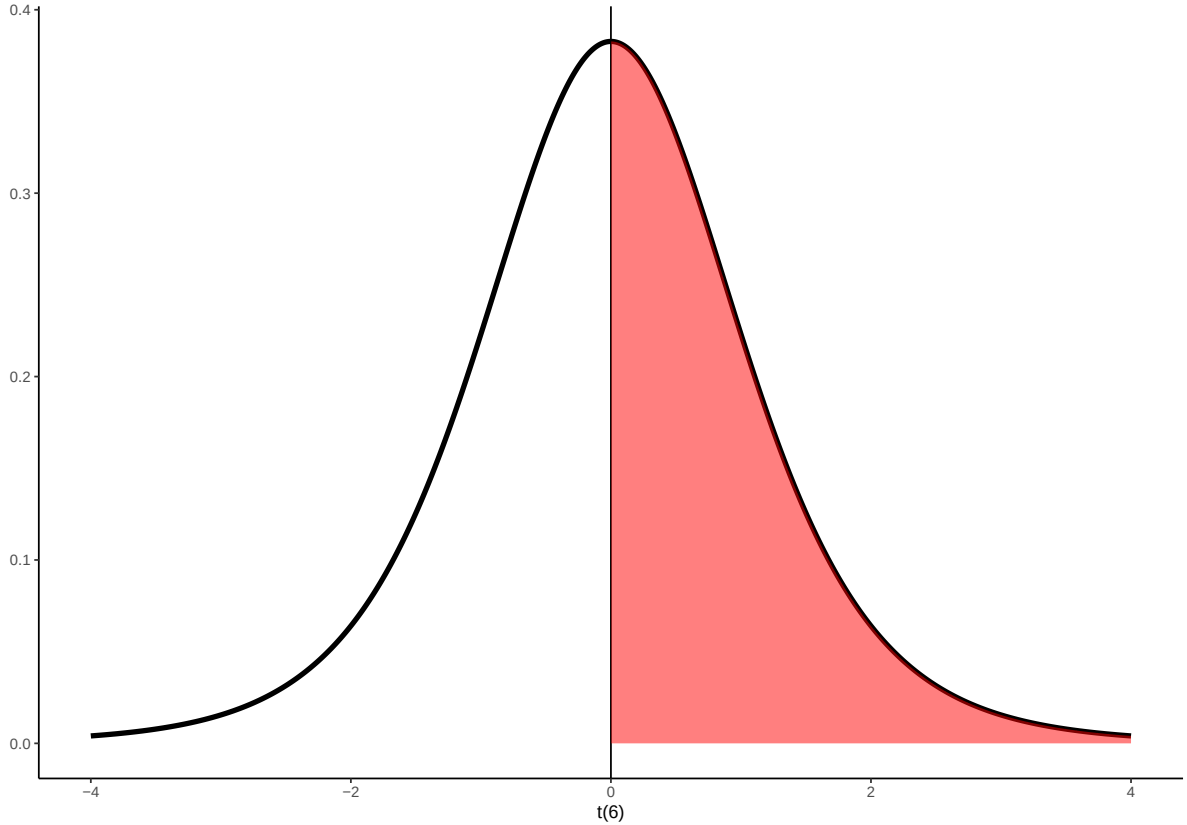

Question 1(d)

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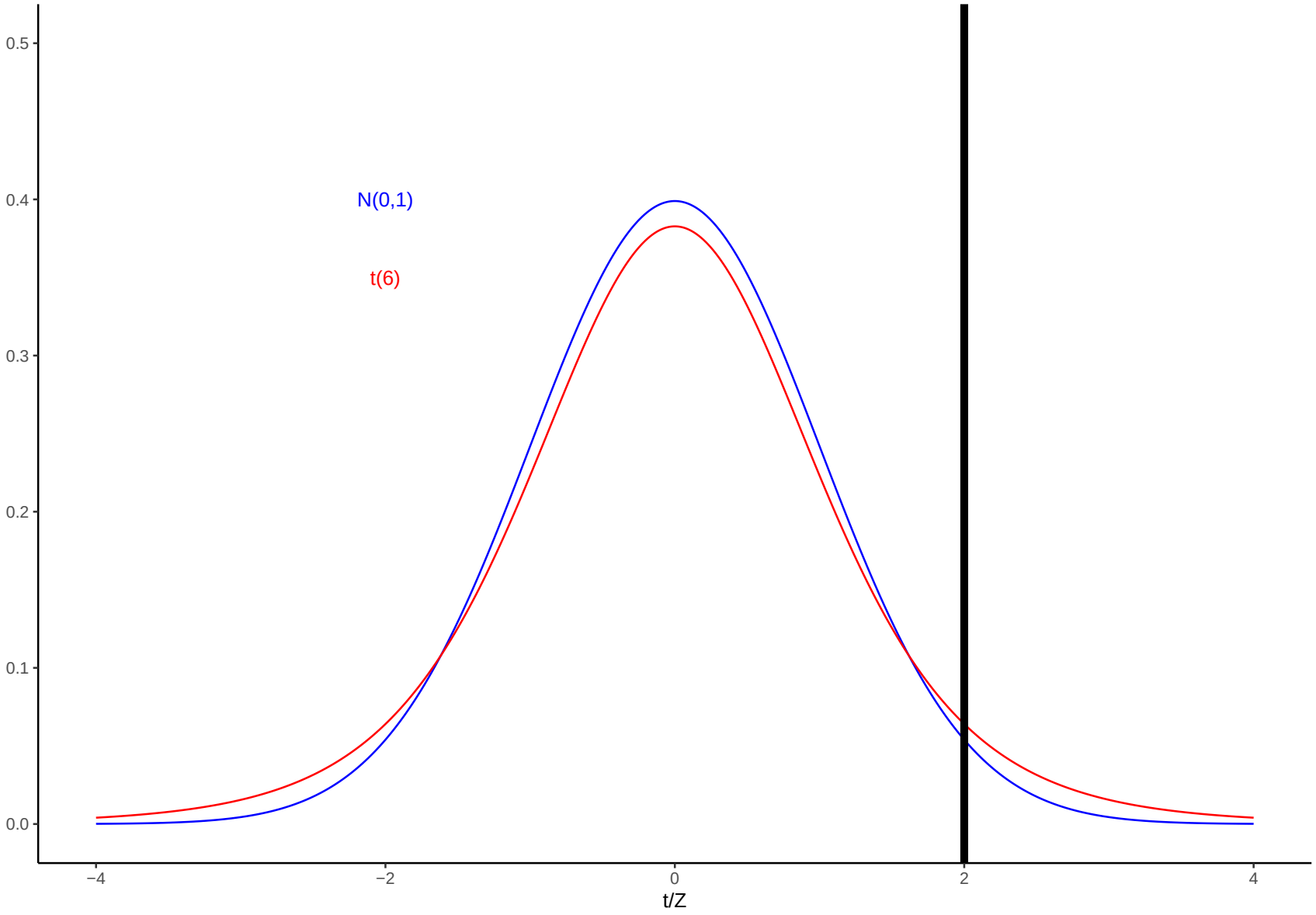
- Recall part (b) that $V(t)$ is greater than 1. So the t distribution with $\nu = 6$ has a higher variance than the standard normal distribution.
- The area under the probability density function to the right of 2 would be greater for the t distribution.
- In other words, the extreme values (values further from the mean of 0) are more likely under the t distribution than under the standard normal distribution.

Tail Comparison



Question 2

The outstanding balances at the end of each calendar month on credit cards issued by a large bank are known to be normally distributed with variance $\$623^2$.

A sample of 100 outstanding balances at the end of March has a mean of $\$2,815$.

a. How would you construct a 95% confidence interval estimate for the mean outstanding balance?

b. Does your answer depend on the information that the balances are normally distributed?

c. Would your answer depend on the information that the balances are normally distributed if the sample size had been 20 rather than 100?

Question 2 - Analysis

Define X to be the outstanding balance of a randomly selected credit card at the end of each calendar month.

- $X \sim N(\mu, \sigma^2 = \$623^2)$

Sample information: $n = 100, \bar{x} = \$2,815$

(a) 95% Confidence Interval

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(a) 95% Confidence Interval

Since σ is known, we use $\bar{x} \pm z_{0.025} \frac{\sigma}{\sqrt{n}}$

CI estimate:

$$2,815 \pm (1.96 \times 62.3) = (\$2,802.79, \$2,827.21)$$

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$$2,815 \pm (1.96 \times 62.3) = (\$2,802.79, \$2,827.21)$$

Interpretation: We are 95% confident that this interval contains the mean outstanding balances at the end of each calendar month on credit cards issued by a large bank.

Question 2 - Discussion

(b) Normality Assumption for $n = 100$

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No. For a sample size of 100, we could invoke the **Central Limit Theorem** and say that the sample mean was approximately normally distributed. Information for the population variance is also known.

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Question 2 - Discussion

(b) Normality Assumption for $n = 100$

No. For a sample size of 100, we could invoke the **Central Limit Theorem** and say that the sample mean was approximately normally distributed. Information for the population variance is also known.

(c) Normality Assumption for $n = 20$

- With a small sample size of 20, we could not invoke the Central Limit Theorem.
- Therefore, without the population being normally distributed, we could not construct a CI estimate using the standard normal distribution.

Question 3

Many people eat salmon due to its rich taste and health benefits. However, there is some concern about the level of contaminants in farmed salmon. There is a recommendation that the average concentration of the cancer-causing insecticide mirex should be no larger than 0.08ppm (parts per million). A purchase manager for a supermarket wants to be confident that the fish they purchase are safe for their customers. They ask you to find the 95% confidence interval that contains the true value of the mean mirex concentration and to interpret it for them. The data they have collected on the testing of 150 farm-raised salmon for their level of mirex concentration is in the file ‘salmon.xlsx‘.

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Which confidence interval should we use?

$$\bar{x} \pm t_{0.025,149} \times \frac{s}{\sqrt{n}}$$

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Summary from Excel:

- $\bar{x} = 0.091$
- $s = 0.0493$
- $n = 150$
- $t_{0.025,149} = 1.976$

Question: Do we have any reason to believe that the true value of the mean mirex concentration exceeds the recommended level of 0.08 ppm?

Question 4

A manufacturer of a brand of designer jeans realizes that many retailers charge less than the suggested retail price of \$80. A random sample of 20 retailers reveals that the mean and the standard deviation of the prices of the jeans are \$64 and \$5 respectively. Estimate with 90% confidence the mean retail price of the jeans, assuming that the prices of jeans are normally distributed.

Question 4

Since $X \sim$ Normal, σ is unknown, **use** t **with** $\mathrm{df} = 19$.

$$t_{\alpha/2,n-1} = t_{0.05,19} = 1.729$$

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$$t_{\alpha/2,n-1} = t_{0.05,19} = 1.729$$

Therefore, 90%CI for μ is

$$\bar{X} \pm t_{\alpha/2,n-1}\left(\frac{s}{\sqrt{n}}\right) = 64 \pm 1.729\left(\frac{5}{\sqrt{20}}\right) = 64 \pm 1.93$$

$$LCL = 62.07, UCL = 65.93$$

Thus, we are 90% confident that this interval contains the mean retail price of the jeans. Because the interval from \$62.07 to \$65.93 is entirely below the recommended retail price (\$80) we have reason to believe that the true value of the mean retail price of the jeans is below the recommended price.

Question 5

In March 2013, workers in a greeting card company in France took their bosses hostage (https://www.rfi.fr/en/economy/20130329-greeetings-card-workers-kidnapbosses-over-unpaid-layoff-pay). Company chiefs had informed employees who were to be laid off that they would not receive the redundancy pay that they were legally entitled to. The workers refused to let their bosses leave the company. There had been a number of similar "bossnappings" in 2009 in France. A poll taken by Paris Match (a weekly news magazine) in April 2009 found 30% of the French "approving" and 63% "understanding" or "sympathetic" of the action. Only 7% condemned the practice of "bossnapping". The poll was based on a random representative sample of 1,010 adults. What can we conclude about the proportion of all French adults who are "understanding" or "sympathetic" of the practice of bossnapping?

Question 5

Define p as the proportion of ALL French adults who are "understanding" or "sympathetic" of the practice of bossnapping.

- $n = 1,010$
- $\hat{p} =$

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Question 5

- $n = 1,010$
- $\hat{p} = 0.63$

Given $n\hat{p} = 1010 \times 0.63 = 636.3 > 5$ and $n\hat{q} = 1010 \times 0.37 = 373.3 > 5$, the approximation that the sample proportion is approximately normal is reasonable.

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Question 5

CI estimate:

$$0.63 \pm \left(1.96 \times \sqrt{\frac{0.63 \times 0.37}{1010}}\right) = 0.63 \pm (1.96 \times 0.015) = (0.601, 0.659)$$

Question 5

CI estimate:

$$0.63 \pm \left(1.96 \times \sqrt{\frac{0.63 \times 0.37}{1010}}\right) = 0.63 \pm (1.96 \times 0.015) = (0.601, 0.659)$$

We are 95% confident that between 60.1% and 65.9% of all French adults were "understanding" or "sympathetic" of the practice of bossnapping.

- This result might be useful for guiding future corporate public relations.

Thanks for your attention!

Let me know if there is any questions!